

Jatin Agrawal

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EDUCATION

Indraprastha Institute of Information Technology Delhi (IIIT-D) M.Tech, Computer Science and Engineering	2025 – Present CGPA: 7.9 (Current)
Aligarh Muslim University (AMU) B.Tech, Computer Science and Engineering	2021 – 2025 CGPA: 8.49
A.M.U. City School Class XII (CBSE)	2019 – 2020 91%

TECHNICAL SKILLS

Languages	C++, C, Python, SQL
CS Fundamentals	Data Structures and Algorithms (DSA), Object-Oriented Programming (OOP), Operating Systems, Database Management Systems (DBMS)
Systems & Infra	Linux (Ubuntu), Docker, Kubernetes, Kubernetes Networking, Container Network Interface (CNI), Git, GitHub
AI / ML	Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangChain, ChromaDB, Whisper, Mistral AI, Streamlit, FFmpeg
Robotics	Robot Operating System 2 (ROS 2), RViz2

PROJECTS

- 1. Autonomous Robot Navigation & Simulation System** Jan 2026 – May 2026
Guide: Dr. Tanmoy Kundu, CSE Faculty, IIIT-D Team Size: 4 | Tools: ROS 2, RViz2, Python
GitHub: github.com/jatin25027/ros_deployed
 - Engineered a decentralized **ROS 2** multi-agent simulation supporting up to **10 autonomous quadcopters** with realistic **PD-controller-based** flight dynamics and distributed **inter-robot communication** across independent agent nodes.
 - Implemented **BFS-based path planning** and dynamic obstacle rerouting with a cell-registry-based **collision and dead-lock avoidance** system to enable safe and efficient multi-drone navigation in shared airspace.
 - Developed a **Finite State Machine (FSM)**-driven autonomous flight pipeline covering takeoff, hovering, navigation, and landing phases, with pitch/roll dynamics and propeller animation for realistic quadcopter behavior.
 - Integrated real-time **RViz2 visualization** displaying live drone trajectories, waypoints, and system states, enabling interactive monitoring and debugging of the multi-agent simulation.
- 2. Performance Comparison of Kubernetes CNI Plugins** Jan 2026 – Jun 2026
Guide: Dr. Rinku Shah, CSE Faculty, IIIT-D Team Size: 4 | Tools: Kubernetes, Docker, Python, Shell, kind
GitHub: github.com/jatin25027/cni_plugin
 - Built an automated benchmarking framework to evaluate Kubernetes **Container Network Interface (CNI) plugins** — **Flannel, Calico, and Cilium** — across distributed Kubernetes clusters provisioned using **kind**.
 - Automated **latency, throughput, and resource utilization** benchmarking using Python and Shell scripting, and developed a **performance analysis pipeline** to collect, compare, and visualize results across CNI plugins for evaluating **network efficiency and resource overhead**.
 - Analyzed **eBPF-based** data plane architecture in **Cilium** versus traditional overlay networking in Flannel and Calico, quantifying the impact of kernel-level packet processing on cluster network performance.
 - Demonstrated that **Cilium reduced p95 latency by approximately 33%** (2.73 ms → 1.83 ms) and increased **throughput by approximately 22%** (855 → 1045 RPS) compared to Flannel, providing actionable insights for Kubernetes networking decisions in production environments.
- 3. AI Video Summarizer & Meeting Assistant** Mar 2026 – Present
Tools: Python, Streamlit, LangChain, ChromaDB, Mistral AI, Whisper, FFmpeg, Pydub Team Size: 1
GitHub: github.com/jatin25027/ai-video-assistant
 - Built an **AI-powered Video Assistant** using **Streamlit** and **LangChain** to automate transcription, summarization, and insight extraction from video files, reducing manual meeting review time by up to **80%**.
 - Implemented a scalable, dual-engine **speech-to-text (STT)** pipeline integrating **Whisper** (English) and **Sarvam AI**

(Hinglish), capable of processing **60+ minute** audio files using **FFmpeg** and **Pydub** chunking to overcome API and memory limits.

- Engineered a custom structured extraction module using **Mistral AI LLM** to automatically parse and categorize meeting transcripts into **Video Chapters, Action Items, Key Decisions, and Open Questions** for structured meeting analytics.
- Developed a **Retrieval-Augmented Generation (RAG)** conversational interface backed by **ChromaDB** vector storage, enabling high-accuracy **semantic search** and natural language Q&A across lengthy meeting transcripts using embedding-based retrieval.
- Integrated a dynamic **Knowledge Graph visualization** to map key entities (Organizations, People, Technologies) and their inter-relationships extracted directly from transcript data, providing structured meeting intelligence at a glance.

EXPERIENCE / POSITIONS OF RESPONSIBILITY

Teaching Assistant — Data Structures and Algorithms

Jan 2026 – May 2026

Department of Computer Science and Engineering, IIIT Delhi

- Conducted weekly tutorials and lab sessions on core DSA topics including **trees, graphs, dynamic programming, sorting, and searching algorithms** for **60+ undergraduate students**.
- Designed and evaluated assignments, quizzes, and exam questions in coordination with course faculty; provided structured feedback on code correctness and complexity analysis.
- Held regular office hours to debug student code and clarify algorithmic concepts, directly supporting student understanding and improving assignment completion rates.

Teaching Assistant — Programming in Python

Aug 2025 – Dec 2025

Department of Computer Science and Engineering, IIIT Delhi

- Mentored **30+ first-year students** in Python fundamentals through weekly lab sessions covering data structures, **Object-Oriented Programming (OOP)**, and file handling.
- Reviewed and graded weekly programming assignments; identified common bugs and logic errors for in-class discussion and provided code-review guidance during lab hours.
- Assisted in preparing lab exercises and sample solutions aligned with course learning outcomes and curriculum standards.

ACHIEVEMENTS

- **GATE 2025:** Achieved a score of **661**; ranked in the **Top 1%** among approximately 170,000 candidates. Also qualified **GATE 2024**.
- **Competitive Programming:** Solved **500+ Data Structures and Algorithms (DSA) problems** across competitive programming platforms including **LeetCode, GeeksforGeeks (GFG), and Codeforces**.
- **Coding Expert:** Selected as a Coding Expert at Codingal **Top 1%** of applicants to clear all stages of the selection process.
- **AMUEEE:** Secured **All India Rank (AIR) 95** in the Aligarh Muslim University Engineering Entrance Examination (AMUEEE).
- **JEE Main:** Achieved **99.7+ percentile in Mathematics** in the Joint Entrance Examination (JEE) Mains.